J.P.Morgan

US Natural Gas, NGLs, and Power

Tariff reality check for energy

Global Commodities Research

Shikha Chaturvedi
(1-212) 834-3245
shikha.chaturvedi@jpmorgan.com
JPMorgan Chase Bank NA

- **Effective Tuesday, February 4th, the US will impose a 25% tariff on goods from Mexico and Canada, with a reduced 10% levy on Canadian energy, and an additional 10% tariff on all imports from China.**
- **Much could change between now and Tuesday and duration of the tariffs matters immensely to the ultimate economic and price repercussions.**
- **A more supportive Henry Hub gas price will be seen initially, holding all else constant. However, we would view it as relatively minimal with the brunt of the impact likely felt in more depressed Canadian prices while regional prices in the US, particularly in the West, find some upward pressure to reflect a roughly $.20-.25/MMBtu impact from a 10% tariff. Much of that has been reflected in pricing relationships already. At the moment, we do not expect a change in net Canadian natural gas imports as a result of these tariffs during summer 2025.**
- **Given the anticipated growth in US propane supply, particularly from PADD 3, and as a reaction to US import tariffs, we would expect not only a reduction in Canadian propane imports to the US but also further downward pressure on Edmonton propane prices as has already been evident since the start of the year.**
- **A potential reduction in US electricity imports from Canada could manifest, even through retaliation, but would come with higher power prices for the US regions impacted, particularly NYISO and ISO-NE.**
- **Retaliatory measures are likely to cause redirection of flows for both LNG and propane, particularly US exports to China. Mexico is unlikely to shift its import profile of US natural gas and propane without paying a significant premium for LNG and Middle Eastern propane, though US exports could diminish given the expected deleterious impact to Mexico's economy. Less flexibility in any retaliation to US ethane exports by China and/or Canada could find increased ethane rejection in the US natural gas stream.**

See page 12 for analyst certification and important disclosures.

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

Shikha Chaturvedi [AC]
(1-212) 834-3245
shikha.chaturvedi@jpmorgan.com

Effective Tuesday, February 4th, the US will impose a 25% tariff on goods from Mexico and Canada, with a reduced 10% levy on Canadian energy, and an additional 10% tariff on all imports from China. The tariffs will be enacted under the authority of the IEPPA, with illegal immigration and drug trafficking declared as national emergencies. The duties will remain in effect until the White House is satisfied that the drug and migrant crisis at US borders is "alleviated". Notably, the tariffs include a retaliation clause that will increase penalties if the trading partners respond with their own tariffs against the US.

The administration did not immediately specify the tariff codes covered by the 10% energy duty. However, "energy" or "energy resources" are defined by the Mineral Security Law 30 U.S.C. 1606 (a)(3) to include crude oil, natural gas, lease condensates, natural gas liquids, refined petroleum products, uranium, coal, biofuels, geothermal heat, and critical minerals.

In a statement issued Saturday evening, Canada has announced plans for retaliatory tariffs on the US, imposing 25% counter-tariffs on $155bn worth of American-made products—$30bn of which will take effect by Tuesday, with the remaining $120bn following in 21 days. Mexico has also unveiled a plan that includes tariffs and non-tariff measures to defend its interests, though it has not specified which US goods will be targeted.

Precedent set, precedent followed

As we pointed out in our US NGLs note last week (see here), the actions by the Trump administration with Colombia last weekend created precedent for energy inclusion for US import tariffs. The signing of executive orders this weekend revealed that this precedent was going to be followed for the time being across the energy sector.

Our perspective has been that imposing tariffs on energy imports is inconsistent with an administration whose primary policy goal is to maintain low energy and consumer prices. Consequently, we assumed that crude and natural gas will likely be exempt from any tariffs, and even if such tariffs are imposed, they will likely be short-lived (*Trump's tariff toolbox*, 17 January 2025). While the outlook is unclear, Trump's decision not to exclude energy from the tariffs on Canada clearly goes against our logic and demonstrates his willingness to accept economic risks to pressure other nations into complying with his policy demands.

Much could change between now and Tuesday and duration of the tariffs matters immensely to the ultimate economic and price repercussions. With this in mind, we lay out our initial thoughts on natural gas, NGLs, and power fundamental impacts below as well as directional price views.

Tariffs to impact natural gas, LPG, and electricity imports

Below is a summary of impacts to US energy imports aside from crude oil and condensate. The impacts to the crude and condensate sector can be found here. (see *Oil Markets Weekly: Monroe doctrine revival: neutral now, potential to turn bearish*, Kaneva et al, 2-Feb-25)

US Natural Gas: West balances and price exposed to tariff risk

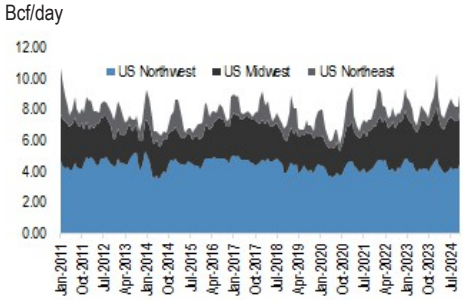
While US imports of natural gas from Mexico are negligible, the US is a net importer of a significant amount of Canadian natural gas. The U.S. Energy Information Administration (EIA) reports that during the first 11 months of 2024, the U.S. net imported 5.7 Bcf/day on average (gross imports = 8.4 Bcf/day). Total U.S. natural gas net imports from Canada

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

Shikha Chaturvedi [AC]
(1-212) 834-3245
shikha.chaturvedi@jpmorgan.com

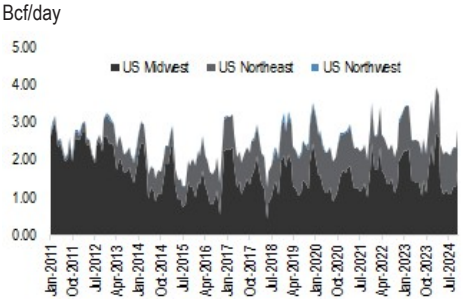
accounted for 5.3% off supply feeding U.S. demand for the same time period in 2024. The largest import points for the U.S. during 2024 were through Eastport, Idaho and Sumas, Washington feeding the western US natural gas balance and accounting for 44% of gross imports reported for 2024. The second largest import points reside at the Sherwood, North Dakota, and Noyes, Minnesota border points and feed the US Midwest natural gas markets, accounting for 32% of gross imports reported for 2024. With the US also exporting natural gas to Canada through the Midwest and the Northeast, the primary impacted US region from a net import perspective is the Pacific Northwest.

Figure 1: US imports of Canadian NG



Source: EIA, J.P. Morgan Commodities Research

Figure 2: US NG exports to Canada



Source: EIA, J.P. Morgan Commodities Research

A more supportive Henry Hub gas price will manifest initially. However, we would view it as relatively minimal price move with the brunt of the impact likely felt in more depressed Canadian prices while regional prices in the US, particularly in the West, find some upward pressure to reflect a roughly $.20-.25/MMBtu impact from a 10% tariff. Much of that has been reflected in pricing relationships already. Within the US, a potential call on increased pipeline flows from East to West, through underutilized pipelines such as Ruby Pipeline, could help to redirect and increase gas flows to the Rockies with regional spreads dictating this. It is important to note that a couple of years of low natural gas prices has stunted gas growth within the region, and any reduction of Canadian imports to the Pacific Northwest or even the Midwest is likely to see outsized regional gas price moves in the US, particularly as the US gas balance is unlikely to see a strong production response in these regions at least for 2025. This year, we currently assume nearly flat net Canadian imports to the US yoy, but we will have to monitor the evolution and longevity of tariffs on energy before we revise this forecast.

US LPGs: Domestic supply offers a solution

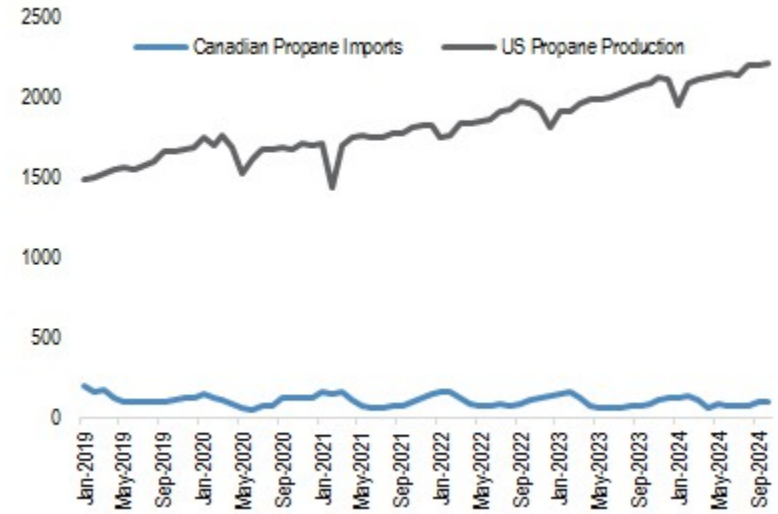
Published late-last week (see here), US import tariffs impact a significantly smaller amount of propane and butane imports from Canada relative to the crude or gas trade. During the first 11 months of 2024, the U.S. imported an average of 150 kbd of these liquefied petroleum gases (LPGs) from Canada, with propane and normal butane accounting for 66% and 27%, respectively, of the total. During this period, Canadian propane exports to the U.S. averaged nearly 100 kbd and contrasts greatly with U.S. domestic supply averaging 2.16 mbd. Given continued robust growth anticipated for US domestic propane production, it is likely that the US will be able to reduce its reliance on Canadian LPG imports this summer. We estimate that by 2026, U.S. propane production will average 2.3 mbd. A similar trend is anticipated for butane, with domestic supply growth paralleling that of propane.

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

Shikha Chaturvedi [AC]
(1-212) 834-3245
shikha.chaturvedi@jpmorgan.com

Figure 3: Canadian propane imports vs US propane production

kbd

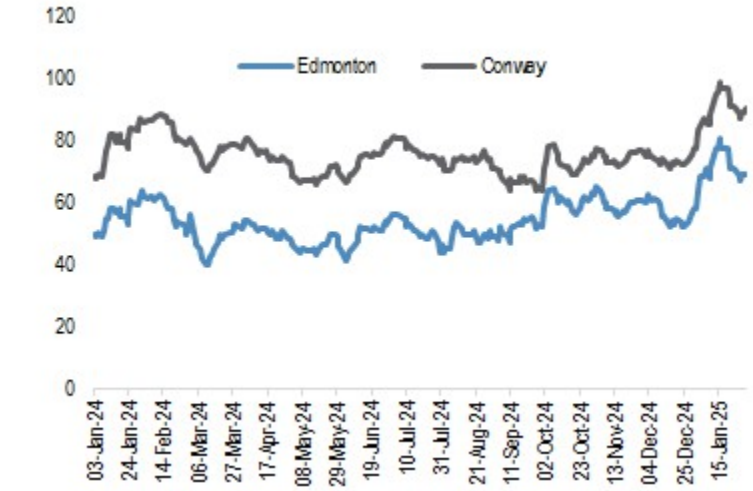


Source: EIA, J.P. Morgan Commodities Research

Due to the seasonal nature of U.S. propane demand, which peaks in winter given increased heating needs, this tariff on Canadian LPG imports could have more pronounced price implications with the remainder of winter weather still uncertain. This is especially true given that U.S. propane stocks are currently experiencing a significant drawdown, with January weather realizing ~1.5 standard deviations colder than the 10-year normal, according to CWG. However, given the anticipated growth in U.S. supply, particularly from PADD 3, and as a reaction to US import tariffs, we would expect not only a reduction in Canadian imports to the U.S. but also downward pressure on Edmonton propane prices. This price movement is already evident in the decline in Edmonton propane prices relative to Conway since the start of the year. While we do not explicitly publish butane balances, we believe a similar dynamic will occur for Canadian butane imports.

Figure 4: Edmonton vs Conway price spread

cpg



Source: Bloomberg Finance L.P., J.P. Morgan Commodities Research

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

Shikha Chaturvedi [AC]
(1-212) 834-3245
shikha.chaturvedi@jpmorgan.com

US Electricity: Low imports but likely impactful to specific US regional prices

According to EIA, US imports of electricity from Canada and Mexico in 2023 reached 38.9 TWh, accounting for less than 1% of total US net generation. With 85% of those imports originating in Canada, while overall import levels are low, import tariffs are most likely to impact New York Independent System Operator (NYISO) and ISO- New England (ISO-NE) where increased reliance of electricity imports is present from a state perspective. S&P Global reports that New York state imported 7.7 TWh in 2024, while Vermont and Maine imported 3.9 Twh and 2.3 TWh, respectively. Additionally, cross border electricity flows from British Columbia, Canada to the US West could be impacted, with S&P Global reporting that "the premiers of two Canadian power-exporting provinces, Doug Ford of Ontario and David Eby of British Columbia, threatened to cut exports to the U.S. if the tariffs were imposed." As such, a potential reduction in U.S. electricity imports could manifest but would come with higher power prices for the regions impacted.

Retaliatory tools to force a redirection of NG and NGLs flows

We covered in detail the retaliation tools held by Canada, Mexico, and China potentially impacting US natural gas and NGLs exports (see *Global Commodities: Trump's tariff toolbox*, Kaneva, 17-Jan-2025 and *US NGLs: More likely to get caught in tariff war crosshairs than crude and gas*) and summarize them below.

Mexico: We would expect that any retaliation by Mexico, potentially reducing propane or natural gas imports, would be extremely difficult given the significantly higher cost to import LNG or Middle Eastern propane. Though, a more meaningful reduction in Mexican exports of natural gas and propane could manifest with the potential deleterious impact that these tariffs may have to Mexico's economy.

Canada: As it pertains to Canada, impacts of flows of U.S. ethane and natural gasoline exports could manifest, though the latter is necessary for oil sands production. A reduction of US ethane exports to Canada could result in increased ethane rejection in the U.S. natural gas stream.

China: As the possibility of U.S. import tariffs on Chinese goods looms, it's important to note that the U.S. has been expanding its export profile to China over the past several years, not only through liquefied natural gas (LNG) but also with NGLs, particularly ethane and propane. If China were to reduce its imports of U.S. LNG, a diverse global consumer base is likely to absorb U.S. natural gas. A similar scenario could unfold for propane, with China importing approximately 220 kbd of U.S. propane in 2023. Increased propane imports from the Middle East could allow China to maintain its rapid pace of PDH plant expansions.

Ethane, however, presents less flexibility for both China and the U.S. U.S. ethane exports to China have steadily increased, averaging 212 kbd in 2023 and 227 kbd in the first ten months of 2024, accounting for 47% of U.S. ethane exports. S&P Global Commodity Insights projects significant growth in China's ethane-based ethylene production, with 18.3 mmt of capacity additions expected from 2025 to 2028, on top of the 26 mmt already in place by the end of 2024. This growth could increase China's demand for ethane feedstock by approximately 1.1 mbd, most of which would likely need to come from the U.S., the world's largest ethane producer.

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

However, in a scenario involving U.S. import tariffs and retaliatory measures, if ethane becomes a target, it could slow both ethylene growth in China and U.S. ethane exports to the country. Given the limited supply sources of ethane, a complete reduction in U.S. ethane exports to China as a retaliatory measure seems unlikely. However, even halving the current pace of ethane exports could result in more U.S. ethane being redirected into the natural gas stream.

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

Shikha Chaturvedi [AC] (1-212) 834-3245
shikha.chaturvedi@jpmorgan.com
JPMorgan Chase Bank NA

Figure 5: J.P. Morgan Commodities 2024 Lower 48 natural gas balance

	2024F														2024F			2024F Win 23-24	Sum 24
Bcf/day	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	Year		
Total Supply	109.9	109.9	105.6	105.7	106.2	107.9	110.0	108.5	106.8	107.9	108.9	111.0	108.5	106.6	108.4	109.3	108.2	109.1	107.6
Dry Gas Production	102.5	104.9	101.6	100.7	100.6	101.9	103.3	102.2	101.0	102.0	102.6	104.6	103.0	101.1	102.2	103.1	102.3	103.6	101.7
Net Canadian Imports	7.3	4.9	4.0	5.0	5.5	6.0	6.6	6.3	5.8	5.9	6.3	6.3	5.4	5.5	6.2	6.2	5.8	5.4	5.9
LNG Imports	0.1	0.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.1	0.0
Total Domestic Consumption	117.1	98.8	89.7	77.7	74.3	80.2	87.8	86.1	79.8	77.1	90.0	107.3	101.9	77.4	84.6	91.5	88.8	99.2	80.4
Power Generation	36.6	31.5	29.7	29.3	33.7	41.4	49.3	48.0	40.9	33.5	32.6	33.8	32.6	34.8	46.0	33.3	36.7	32.5	39.4
Industrial	25.1	24.2	23.9	22.7	21.9	21.5	21.7	21.8	22.2	22.2	23.9	25.4	24.4	22.0	21.9	23.8	23.0	24.3	22.0
Residential/Commercial	45.4	33.9	27.3	17.3	10.6	8.9	7.9	7.8	8.6	13.0	25.0	38.7	35.5	12.2	8.1	25.6	20.4	33.2	10.6
Lease, Plant, and Fuel	9.9	9.2	8.8	8.4	8.1	8.4	8.8	8.5	8.2	8.4	8.5	9.4	9.3	8.3	8.5	8.8	8.7	9.2	8.4
Total Exports*	19.8	19.6	19.0	18.3	19.3	19.5	18.9	19.7	19.6	19.1	19.4	19.8	19.4	19.0	19.4	19.4	19.3	19.7	19.2
Pipeline to Mexico	6.0	5.9	5.9	6.4	6.5	6.8	7.0	6.9	6.9	6.5	5.8	5.6	5.9	6.6	6.9	6.0	6.3	5.8	6.7
Feedgas for LNG exports	13.8	13.7	13.1	11.9	12.8	12.7	11.9	12.8	12.7	12.6	13.6	14.2	13.5	12.5	12.5	13.5	13.0	13.9	12.5
LNG Exports	12.8	12.4	11.9	10.1	11.9	11.9	10.4	11.7	12.1	12.1	12.3	12.8	12.4	11.3	11.4	12.4	11.9	12.7	11.5
Net Injection to storage**	-27.0	-8.5	-3.1	9.8	12.5	8.2	3.3	2.7	7.4	11.8	-0.6	-16.1	-12.8	10.2	4.4	-1.7	0.0	-9.8	8.0
Net Injection to storage (Bcf)**	-836	-245	-95	294	389	247	101	83	222	364	-19	-500	-1168	927	409	-152	12	-1492	1701
End of period storage level (Bcf)	2569	2323	2229	2522	2911	3158	3259	3342	3564	3929	3910	3409	2229	3158	3564	3409	3409	2229	3929

Note: * Total exports do not include exports to Canada. Pipeline exports to Canada are included in Net Canadian Imports in the Total Supply section. ** Withdrawals represented as a negative number.

Source: JPMorgan Natural Gas Strategy, EIA, corporate reports

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

Shikha Chaturvedi [AC] (1-212) 834-3245
shikha.chaturvedi@jpmorgan.com
JPMorgan Chase Bank NA

Figure 6: J.P. Morgan Commodities 2025 Lower 48 natural gas balance

	2025F													2025F				2025F Win 24-25	Sum 25
Bcf/day	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	Year		
Total Supply	112.0	112.2	110.9	110.0	110.1	110.4	111.1	111.3	111.4	112.2	113.6	114.0	111.7	110.2	111.3	113.3	111.6	111.0	110.9
Dry Gas Production	103.9	105.0	104.9	104.3	104.4	104.7	105.1	105.3	105.7	106.2	107.6	107.9	104.6	104.4	105.4	107.3	105.4	104.2	105.1
Net Canadian Imports	8.0	7.0	6.0	5.7	5.7	5.7	5.9	6.0	5.7	6.0	6.0	6.0	7.0	5.7	5.9	6.0	6.1	6.7	5.8
LNG Imports	0.1	0.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.1	0.0
Total Domestic Consumption	123.6	109.5	93.4	79.7	73.5	78.8	86.5	85.8	78.8	78.2	92.0	105.0	108.8	77.3	83.7	91.7	90.4	104.8	80.2
Power Generation	36.7	32.4	29.2	28.1	31.1	39.5	47.7	46.9	39.4	33.1	31.0	31.8	32.8	32.9	44.6	32.0	35.6	32.9	38.0
Industrial	27.0	26.1	24.6	23.5	22.3	22.1	22.1	22.5	22.6	23.0	25.1	25.8	25.9	22.6	22.4	24.6	23.9	25.4	22.6
Residential/Commercial	49.6	41.7	30.8	19.7	12.0	8.9	8.0	7.8	8.6	13.7	27.4	37.9	40.7	13.5	8.2	26.4	22.2	37.2	11.2
Lease, Plant, and Fuel	10.3	9.2	8.8	8.4	8.1	8.4	8.8	8.5	8.2	8.4	8.5	9.4	9.4	8.3	8.5	8.8	8.8	9.3	8.4
Total Exports*	20.3	21.2	20.8	20.6	21.0	21.3	20.8	22.2	22.2	22.4	22.5	23.1	20.8	21.0	21.7	22.7	21.5	20.3	21.5
Pipeline to Mexico	5.7	6.0	6.0	6.4	6.5	6.8	7.0	7.1	7.1	6.7	5.8	5.6	5.9	6.6	7.1	6.0	6.4	5.8	6.8
Feedgas for LNG exports	14.6	15.2	14.8	14.2	14.5	14.5	13.8	15.1	15.2	15.7	16.7	17.5	14.9	14.4	14.7	16.6	15.1	14.5	14.7
LNG Exports	13.1	13.7	13.3	12.8	13.0	13.1	12.4	13.6	13.6	14.1	15.1	15.7	13.4	12.9	13.2	15.0	13.6	13.1	13.2
Net Injection to storage**	-31.9	-18.5	-3.2	9.7	15.7	10.2	3.7	3.3	10.4	11.6	-0.9	-14.0	-17.9	11.9	5.8	-1.1	-0.3	-14.1	9.2
Net Injection to storage (Bcf)**	-990	-519	-100	291	486	307	115	101	313	360	-27	-435	-1612	1080	534	-100	-121	-2128	1977
End of period storage level (Bcf)	2420	1901	1801	2092	2578	2885	3000	3101	3415	3775	3748	3313	1801	2885	3415	3313	3313	1801	3775

Note: * Total exports do not include exports to Canada. Pipeline exports to Canada are included in Net Canadian Imports in the Total Supply section. ** Withdrawals represented as a negative number.

Source: JPMorgan Natural Gas Strategy, EIA, corporate reports

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

Shikha Chaturvedi [AC] (1-212) 834-3245
shikha.chaturvedi@jpmorgan.com
JPMorgan Chase Bank NA

Figure 7: J.P. Morgan Commodities 2026 Lower 48 natural gas balance

	2026F												2026F				2026F	Win 25-26	Sum 26
Bcf/day	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	Year		
Total Supply	114.3	114.5	114.4	114.3	114.2	114.1	114.0	114.3	114.7	115.2	115.8	116.4	114.4	114.2	114.3	115.8	114.7	114.2	114.4
Dry Gas Production	108.2	108.4	108.4	108.3	108.2	108.0	108.0	108.3	108.7	109.2	109.8	110.3	108.3	108.2	108.3	109.8	108.6	108.1	108.4
Net Canadian Imports	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
LNG Imports	0.1	0.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.1	0.0
Total Domestic Consumption	115.3	109.1	94.6	80.7	74.5	79.7	87.9	87.1	79.9	79.2	92.3	105.2	106.3	78.3	85.0	92.2	90.5	103.2	81.3
Power Generation	32.4	31.7	30.0	28.7	31.8	40.0	48.7	47.9	40.2	33.7	30.9	31.8	31.3	33.5	45.6	32.1	35.6	31.4	38.7
Industrial	27.2	26.3	24.8	23.7	22.5	22.3	22.3	22.7	22.8	23.2	25.3	26.0	26.1	22.8	22.6	24.8	24.1	25.9	22.8
Residential/Commercial	45.1	41.7	30.8	19.7	12.0	8.9	8.0	7.8	8.6	13.7	27.4	37.9	39.2	13.5	8.1	26.3	21.8	36.6	11.2
Lease, Plant, and Fuel	10.5	9.4	9.0	8.6	8.3	8.6	9.0	8.7	8.4	8.6	8.7	9.6	9.6	8.5	8.7	9.0	9.0	9.4	8.6
Total Exports*	23.6	24.2	23.6	23.4	23.7	23.9	23.1	24.5	24.3	24.3	24.3	24.7	23.8	23.7	24.0	24.5	24.0	23.4	23.9
Pipeline to Mexico	6.1	6.4	6.4	6.8	6.9	7.2	7.4	7.5	7.5	7.1	6.2	6.0	6.3	7.0	7.5	6.4	6.8	6.1	7.2
Feedgas for LNG exports	17.5	17.8	17.2	16.6	16.8	16.7	15.7	17.0	16.9	17.2	18.1	18.7	17.5	16.7	16.5	18.0	17.2	17.3	16.7
LNG Exports	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	15.7	15.0	14.8	16.2	15.5	6.2	0.0
Net Injection to storage**	-25	-19	-4	10	16	10	3	3	10	12	-1	-13	-16	12	5	-1	0	-12.4	9.2
Net Injection to storage (Bcf)**	-760	-528	-115	309	494	314	93	82	313	360	-25	-418	-1414	1113	492	-82	28	-1872	1968
End of period storage level (Bcf)	2553	2025	1910	2219	2713	3027	3120	3202	3515	3875	3850	3432	1910	3027	3515	3432	3432	1910	3875

Note: * Total exports do not include exports to Canada. Pipeline exports to Canada are included in Net Canadian Imports in the Total Supply section. ** Withdrawals represented as a negative number.

Source: JPMorgan Natural Gas Strategy, EIA, corporate reports

Figure 8: J.P. Morgan Commodities 2024 US ethane balance

	2024E												2024E				2024E
kbd	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
US Field Production	2,369	2,689	2,840	2,938	2,953	2,862	2,732	2,753	2,911	2,969	2,937	2,903	2,633	2,918	2,799	2,936	2,821
US Refinery and Blend Net Production	11	7	6	11	8	8	10	10	9	9	6	6	8	9	10	7	8
US Imports	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Supply	2,380	2,696	2,846	2,949	2,961	2,870	2,742	2,763	2,920	2,978	2,943	2,909	2,641	2,927	2,808	2,943	2,830
US Exports	505	497	434	465	427	491	480	421	564	504	450	500	479	461	488	485	478
US Products Supplied	2,118	2,364	2,259	2,207	2,355	2,273	2,080	2,387	2,432	2,537	2,421	2,421	2,247	2,278	2,300	2,460	2,321
Total Demand	2,623	2,861	2,693	2,672	2,782	2,764	2,560	2,808	2,996	3,041	2,871	2,921	2,726	2,739	2,788	2,944	2,799
US Stock Change	(243)	(165)	153	277	179	106	182	(45)	(76)	(63)	71	(12)	-85	187	20	-1	30
Net Storage Injection	(7,533)	(4,785)	4,743	8,310	5,549	3,180	5,642	(1,395)	(2,280)	(1,953)	2,139	(375)	-2,525	5,680	656	-63	937
End of Period Storage Balance	56,341	51,556	56,299	64,609	70,158	73,338	78,980	77,585	75,305	73,352	75,491	75,117	56,299	73,338	75,305	75,117	75,117

Note: End of period storage balance in kb
Source: J.P. Morgan Commodities Research, EIA, S&P Global Commodity Insights, Wood Mackenzie, company reports

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

Shikha Chaturvedi [AC] (1-212) 834-3245
shikha.chaturvedi@jpmorgan.com
JPMorgan Chase Bank NA

Figure 9: J.P. Morgan Commodities 2025 US ethane balance

	2025F												2025F				2025F
kbd	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
US Field Production	2,856	2,853	2,846	2,847	2,855	2,869	2,883	2,898	2,913	2,926	2,939	2,950	2,852	2,857	2,898	2,938	2,886
US Refinery and Blend Net Production	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
US Imports	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0
Total Supply	2,862	2,859	2,852	2,853	2,861	2,875	2,889	2,904	2,919	2,932	2,945	2,956	2,858	2,863	2,904	2,944	2,892
US Exports	500	450	400	500	450	560	580	600	600	620	600	620	450	503	593	613	540
US Products Supplied	2,296	2,414	2,309	2,257	2,405	2,323	2,130	2,437	2,482	2,587	2,421	2,421	2,340	2,328	2,350	2,476	2,374
Total Demand	2,796	2,864	2,709	2,757	2,855	2,883	2,710	3,037	3,082	3,207	3,021	3,041	2,790	2,832	2,943	3,090	2,914
US Stock Change	66	(5)	143	96	6	(8)	179	(133)	(163)	(275)	(76)	(86)	156	-61	42	78	-21
Net Storage Injection	2,031	(154)	4,448	2,877	193	(232)	5,563	(4,131)	(4,900)	(8,517)	(2,291)	(2,655)	4,790	-1,876	1,265	2,390	-647
End of Period Storage Balance	77,148	76,994	81,442	84,319	84,511	84,279	89,843	85,712	80,813	72,295	70,004	67,350	81,442	84,279	80,813	67,350	67,350

Note: End of period storage balance in kb
Source: J.P. Morgan Commodities Research, EIA, S&P Global Commodity Insights, Wood Mackenzie, company reports

Figure 10: J.P. Morgan Commodities 2026 US ethane balance

	2026F												2026F				2026F
kbd	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
US Field Production	2,976	2,982	2,980	2,979	2,986	2,999	3,011	3,024	3,036	3,046	3,057	3,066	2,979	2,988	3,024	3,056	3,012
US Refinery and Blend Net Production	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
US Imports	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0
Total Supply	2,982	2,988	2,986	2,985	2,992	3,005	3,017	3,030	3,042	3,052	3,063	3,072	2,985	2,994	3,030	3,062	3,018
US Exports	600	620	600	620	600	620	600	620	600	620	600	620	607	613	607	613	610
US Products Supplied	2,296	2,414	2,309	2,257	2,405	2,323	2,130	2,437	2,482	2,587	2,421	2,535	2,340	2,328	2,350	2,514	2,383
Total Demand	2,896	3,034	2,909	2,877	3,005	2,943	2,730	3,057	3,082	3,207	3,021	3,155	2,946	2,942	2,956	3,128	2,993
US Stock Change	86	(46)	77	108	(13)	62	287	(27)	(40)	(155)	41	(83)	39	52	73	-66	25
Net Storage Injection	2,653	(1,280)	2,375	3,249	(396)	1,854	8,912	(849)	(1,207)	(4,790)	1,240	(2,587)	1,249	1,569	2,285	-2,046	764
End of Period Storage Balance	70,003	68,723	71,097	74,346	73,950	75,804	84,716	83,867	82,660	77,870	79,110	76,523	71,097	75,804	82,660	76,523	76,523

Note: End of period storage balance in kb
Source: J.P. Morgan Commodities Research, EIA, S&P Global Commodity Insights, Wood Mackenzie, company reports

Figure 11: J.P. Morgan Commodities 2024 US propane balance

	2024E												2024E				2024E
kbd	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
US Field Production	1,955	2,093	2,111	2,122	2,137	2,149	2,143	2,200	2,199	2,219	2,231	2,241	2,053	2,136	2,181	2,231	2,150
US Refinery and Blend Net Production	268	253	274	269	278	281	279	287	266	251	256	261	265	276	277	256	269
US Imports	130	156	120	73	87	71	80	85	112	99	123	146	135	77	92	123	107
Total Supply	2,353	2,502	2,505	2,464	2,502	2,501	2,502	2,572	2,577	2,569	2,611	2,648	2,453	2,489	2,550	2,609	2,525
US Exports	1,699	1,848	1,687	1,678	1,683	1,743	1,649	1,847	1,838	1,793	1,850	1,850	1,745	1,701	1,778	1,831	1,764
US Products Supplied	1,285	1,005	759	598	515	480	463	502	613	780	938	1,070	1,016	531	526	929	751
Total Demand	2,984	2,853	2,446	2,276	2,198	2,223	2,112	2,349	2,451	2,573	2,788	2,920	2,761	2,232	2,304	2,760	2,514
US Stock Change	(631)	(351)	59	188	304	278	390	223	126	(4)	(177)	(272)	-308	257	246	-151	11
Net Storage Injection	(19,561)	(10,179)	1,829	5,640	9,424	8,340	12,090	6,913	3,780	(124)	(5,321)	(8,442)	-9,304	7,801	7,594	-4,629	366
End of Period Storage Balance	59,428	49,249	51,078	56,718	66,142	74,482	86,572	93,485	97,265	97,141	91,820	83,377	51,078	74,482	97,265	83,377	83,377

Note: End of period storage balance in kb
Source: J.P. Morgan Commodities Research, EIA, S&P Global Commodity Insights, Wood Mackenzie, company reports

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

Shikha Chaturvedi [AC] (1-212) 834-3245
shikha.chaturvedi@jpmorgan.com
JPMorgan Chase Bank NA

Figure 12: J.P. Morgan Commodities 2025 US propane balance

	2025F												2025F				2025F
kbd	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
US Field Production	2,143	2,241	2,237	2,238	2,243	2,251	2,260	2,268	2,278	2,286	2,293	2,300	2,207	2,244	2,269	2,293	2,253
US Refinery and Blend Net Production	255	236	245	245	253	259	251	254	251	246	255	255	245	252	252	252	250
US Imports	160	156	120	73	87	71	80	85	112	99	123	146	145	77	92	123	109
Total Supply	2,558	2,633	2,602	2,556	2,583	2,581	2,591	2,608	2,641	2,631	2,671	2,701	2,598	2,573	2,613	2,668	2,613
US Exports	1,800	1,800	1,800	1,800	1,800	1,800	1,800	1,850	1,850	1,850	1,850	1,850	1,800	1,800	1,833	1,850	1,821
US Products Supplied	1,397	1,072	836	593	508	486	523	523	641	786	925	1,052	1,102	529	563	921	778
Total Demand	3,197	2,872	2,636	2,393	2,308	2,286	2,323	2,373	2,491	2,636	2,775	2,902	2,902	2,329	2,396	2,771	2,599
US Stock Change	(639)	(238)	(34)	163	275	295	267	234	150	(4)	(104)	(201)	-304	244	217	-103	14
Net Storage Injection	(19,798)	(6,675)	(1,067)	4,883	8,514	8,854	8,284	7,261	4,486	(136)	(3,122)	(6,224)	-9,180	7,417	6,677	-3,161	438
End of Period Storage Balance	63,579	56,904	55,837	60,720	69,233	78,087	86,371	93,631	98,118	97,981	94,859	88,635	55,837	78,087	98,118	88,635	88,635

Note: End of period storage balance in kb
Source: J.P. Morgan Commodities Research, EIA, S&P Global Commodity Insights, Wood Mackenzie, company reports

Figure 13: J.P. Morgan Commodities 2026 US propane balance

	2026F												2026F				2026F
kbd	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
US Field Production	2,307	2,311	2,309	2,309	2,313	2,321	2,328	2,336	2,343	2,349	2,356	2,361	2,309	2,314	2,336	2,355	2,328
US Refinery and Blend Net Production	241	231	240	244	249	254	253	253	247	240	248	248	237	249	251	245	246
US Imports	130	156	120	73	87	71	80	85	112	99	123	146	135	77	92	123	107
Total Supply	2,678	2,698	2,668	2,625	2,649	2,646	2,661	2,674	2,701	2,688	2,727	2,756	2,681	2,640	2,679	2,723	2,681
US Exports	1,900	1,900	1,900	1,900	1,900	1,900	1,900	1,900	1,900	1,900	1,900	1,900	1,900	1,900	1,900	1,900	1,900
US Products Supplied	1,236	1,108	868	631	520	494	549	552	672	749	914	1,051	1,071	548	591	905	779
Total Demand	3,136	3,008	2,768	2,531	2,420	2,394	2,449	2,452	2,572	2,649	2,814	2,951	2,971	2,448	2,491	2,805	2,679
US Stock Change	(458)	(310)	(99)	94	230	252	212	222	129	39	(87)	(195)	-289	192	188	-81	2
Net Storage Injection	(14,202)	(8,685)	(3,082)	2,823	7,118	7,546	6,585	6,874	3,879	1,214	(2,621)	(6,058)	-8,656	5,829	5,779	-2,488	116
End of Period Storage Balance	74,433	65,748	62,666	65,488	72,607	80,153	86,737	93,611	97,490	98,704	96,083	90,025	62,666	80,153	97,490	90,025	90,025

Note: End of period storage balance in kb
Source: J.P. Morgan Commodities Research, EIA, S&P Global Commodity Insights, Wood Mackenzie, company reports

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

Shikha Chaturvedi [AC] (1-212) 834-3245
shikha.chaturvedi@jpmorgan.com
JPMorgan Chase Bank NA

Disclosures

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of http://www.jpmorganmarkets.com where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Company-Specific Disclosures: Important disclosures, including price charts and credit opinion history tables, are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting https://www.jpmm.com/research/disclosures, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to UK MIFID Research Unbundling exemption for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see https://www.jpmorgan.com/disclosures/cryptoasset-disclosure.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit https://www.theocc.com/components/docs/riskstoc.pdf for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or http://www.finra.org/sites/default/files/Security_Futures_Risk_Disclosure_Statement_2018.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below, this material has been distributed by the legal entity responsible for production. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:
Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51). **Australia:** J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting https://www.jpmm.com/research/disclosures. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: J.P. Morgan Australia - Research Independence Policy. **Brazil**: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmorgan.com. **Canada**: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc. **Chile:** Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile. **China:** J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business. **Dubai International Financial Centre (DIFC)**: JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules. **European Economic Area (EEA):** Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons. **Hong Kong**: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited. **India**: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai – 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number – INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: http://www.jpmipl.com. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Spurthi Gadamsetty; spurthi.gadamsetty@jpmchase.com; +912261573225. Grievance Officer: Ramprasadh K, jpmipl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. **Indonesia**: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK). **Korea**: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch. **Japan**: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan. **Malaysia**: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia. **Mexico**: J.P. Morgan Casa de Bolsa, S.A. de C.V. and J.P. Morgan Grupo Financiero are members of the Mexican Stock Exchange and are authorized to act as a broker dealer by the National Banking and Securities Exchange Commission. **New Zealand**: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008. **Philippines**: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission. **Singapore**: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMSS) [MDDI (P) 068/08/2024 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMSS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material. **South Africa**: J.P. Morgan Equities

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA). **Taiwan**: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material. **Thailand**: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission, and its registered address is 3rd Floor, 20 North Sathorn Road, Silom, Bangrak, Bangkok 10500. **UK:** Unless specified to the contrary, research is distributed in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority. JPMS plc is registered in England & Wales No. 2711006, Registered Office 25 Bank Street, London, E14 5JP. This material is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. Research issued by JPMS plc has been prepared in accordance with JPMS plc's policy for prevention and avoidance of conflicts of interest related to the production of Research which can be found at the following link: J.P. Morgan EMEA - Research Independence Policy. **U.S.**: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase &

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.

Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: https://www.jpmorgan.com/disclosures/email

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2025. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to msci.com/disclaimer

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at https://www.sustainalytics.com/legal-disclaimers. ©2025 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised January 01, 2025.

Copyright 2025 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party. It is permissible to use J.P. Morgan Data for internal business purposes only in an AI system or model that protects the confidentiality of J.P. Morgan Data so as to prevent any and all access to or use of such J.P. Morgan Data by any third-party.

Completed 02 Feb 2025 10:15 PM EST

Disseminated 02 Feb 2025 10:15 PM EST

This document is being provided for the exclusive use of Victor Riquelme at Banco Central de Chile.